

On Set 2 (Fig. 2b) containing advanced operations, the two ChatGPT models performed reasonably well, both scoring above 80%, with only a slight drop from the first set. On the other hand, the scores of Gemini, Claude, and Copilot agents drop considerably. These three agents scored between 53-64%, in contrast to their above 85% average scores on the first set.

On Set 3 (Fig. 2c), we examined the agents’ ability to use the knowledge of factors (tested on Set 1) to check if the given numbers are prime. As we can see, the o1 model retained its superior performance level on Set 3. In contrast, the o1-mini agent dropped down to the level of the other three agents, who slightly raised their performance from the advanced operations set.

On Set 4 (Fig. 2d) containing the games of 24, the performance of all LLM agents dropped sharply compared to the first three sets. To provide a comparison, on average, all agents scored between 74% and 95% on the first three sets, whereas the agents scored between 11% and 73% on Set 4. The o1 model is the top performer at 73%, whereas all other agents solved less than 50% of the problems.

These results show that while the agents perform reasonably well in fundamental operations and following deterministic steps to solutions (as reflected in the first three tasks), they struggle to transfer these skills to solve the games of 24. Notably, the 24 game does not require any new operations; rather, the basic operations tested on Set 1 suffice for this game. However, unlike the first three tasks, the 24 game also requires a trial-and-error search for solutions, which has been noted as a challenge for LLMs. In the next section, we examine the mistakes made in each problem type to specify the points of their struggles.

5.3 A Closer Look at the Types of Mistakes

Average LLM performance in each problem is presented in Figure 3. On Set 1, all LLMs can be observed to solve most problems perfectly. The mistakes were mainly limited to loss of precision in decimal fractions. On Set 2, the LLMs struggled with multiple types of operations. All LLMs – barring the ChatGPT agents – struggled with the fractional exponents (such as $400^{0.23}$). The agents also made more mistakes in problems involving unusual bases of logarithms and complex numbers. On Set 3, the LLMs performed generally better than they did on Set 2, but again showed some pitfalls. The most common mistake predictably coincided with large prime numbers. In contrast, they were unaffected by the size of the numbers when dealing with Mersenne primes, but they often mistook the Mersenne Prime Lookalikes (i.e., the non-primes of the form $2^p - 1$) for prime numbers.

On Set 4 containing the games of 24, LLM performance can be observed to fall off a cliff after the initial few problems that were trivially easy and could be solved by simply multiplying the given numbers (such as [1,1,1,24] or [1,1,3,8]). Surprisingly, some agents were not able to solve even these problems. The rest of the problems were more complex than that but still relatively easy for humans (presented in Appendix A), but all LLM agents showed imperfect performance in these problems. A common mistake was assuming a solution did not exist for many of the problems, whereas all problems contained at least one solution. Moreover, the agents frequently broke the rules of the game to achieve 24, mainly in two ways: (1) not using all the numbers and (2) using some numbers

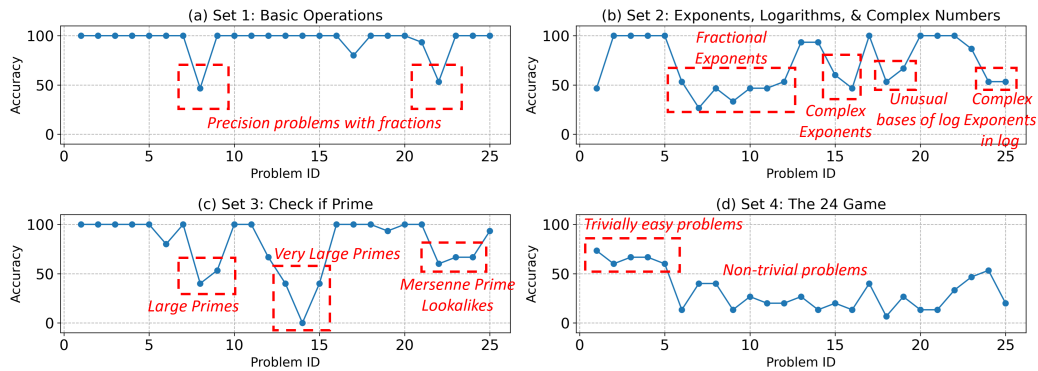


Figure 3: Performance on individual problems. As can be seen, some problems were harder than the rest for the agents. Annotations include problem features as the possible reasons for observed errors.

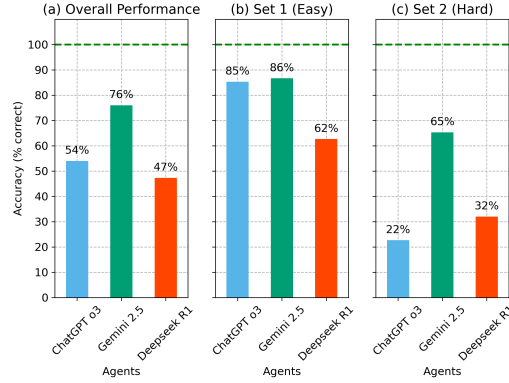


Figure 4: Performance of the three LRM agents in the game of 24 at two levels of difficulty.

more than once. Finally, another common set of mistakes was miscalculating expressions and then providing solutions that do not yield 24 without realizing the miscalculations. These mistakes are particularly surprising as they indicate a breakdown in basic mathematical skills, which we had observed the agents to possess when tested on simpler problems. Notably, the ChatGPT models did not make the last kind of mistakes and performed generally better in playing the 24 game.

5.4 Examining the Effect of Complexity on Reasoning

Figure 4 shows the performance of the three LRM agents in the 24 games. Figure (a) shows the average performance across two levels of difficulty. As we can see, Gemini 2.5 Pro agent performed the best (76%), comfortably outperforming both the ChatGPT o3 (54%) and the Deepseek R1 (47%) agents. Figures (b) and (c) show the performance broken down by problem difficulty level. On Set 1, which is relatively easier, the ChatGPT o3 and the Gemini 2.5 Pro agents perform equally well – 85% and 86% accuracy, respectively – while the Deepseek R1 scored 62%. On Set 2, containing harder games of 24, we see that all agents perform considerably worse compared to the first set. The Gemini agent’s score drops down to 65%, whereas the ChatGPT and the Deepseek agents suffer even larger dips in performance, with their scores going down to 22% and 32%, respectively. These results suggest that the trial-and-error search required in the 24 game is a persistent challenge for modern LLMs, despite enhanced reasoning abilities.

Examining the type of mistakes made, we find that erroneously assuming a solution does not exist remains the most common mistake. Moreover, one such conclusion increased the tendency to conclude the same for later problems. On the other hand, we find that these LRMs were less prone to breaking the rules of the game (e.g., not using all numbers exactly once), with the Gemini model making the fewest such mistakes and the Deepseek model making the most. The frequency of miscalculating expressions also decreased compared to the models tested in Study 1. However, on the harder set, all agents sometimes diverged towards endless chains of reasoning, during which the number and the frequency of miscalculating expressions appeared to increase, indicating a possible effect of compounding errors.

6 General Discussion

As LLM-based agents become our assistants on complex, real-world tasks, understanding their complex reasoning abilities has become essential. In this work, we tested LLM reasoning in the domain of mathematical problem solving as they navigated vast spaces of numbers to find solutions. In our main study, we tested five leading agents on a set of 100 problems across four categories: basic operations, advanced operations, primality checking, and a number game. The agents performed reasonably well in the first three categories, followed by a sharp drop in the number game. This result suggests that while LLMs can effectively execute deterministic problem-solving strategies, they struggle with tasks that require "number sense" – an intuitive understanding of number characteristics and relationships that humans naturally possess and further develop through learning. This limit

becomes evident in the number game, which requires an informed trial-and-error search using knowledge of number factors to guide problem-solving attempts.

More recent tests on new reasoning models indicate that the 24 game and the trial-and-error search inherent in it are persistent challenges for LLMs. While reasoning models perform better on hard problems than non-reasoning models, their improvement comes at the cost of increased computation, as they often embark on long chains of reasoning or code execution. The trade-off between reasoning capability and computational efficiency suggests the need for evaluative frameworks to consider both problem complexity and solution costs for practical applications. Moreover, despite additional computational capabilities, their performance dropped on the harder set of 24 games. Therefore, their superiority over non-reasoning LLMs seems to stem primarily from increases in computational resources rather than improved number senses.

Probing deeper, the types of errors made reflect a fragile number sense in LLMs. Erroneously assuming the non-existence of solutions is the most common mistake, followed by breaking the game rules and miscalculating expressions. An interesting phenomenon we observed is the breakdown of elementary mathematical skills within extended chains of reasoning. While all models included were capable of performing basic operations well, their skills did not always transfer to reasoning chains. They frequently miscalculated simple expressions in producing 24 and, except for the LRMs, were rarely aware of their mistakes.

Taken together, our findings highlight a fragile number sense in LLM agents, despite their skills in following deterministic processes. This limited number sense is likely to make trial-and-error search in vast number spaces a persistent challenge for LLMs. More generally, these results highlight the need for computationally limited agents to be boundedly rational in complex environments, that is, to selectively search for good-enough solutions using heuristics when optimality is beyond reach. As number sense allows humans to solve complex mathematical problems efficiently, we believe it to be essential for improving the efficiency LLM reasoning. The findings from our relatively simple study show the merits of the divide-and-reconstruct approach to decipher how intelligent agents combine the low-level elements of reasoning to solve problems of immense complexity.

Limitations and Future Directions

To establish trial-and-error search in complex environments and, more fundamentally, number sense as limits of LLM mathematical reasoning, they need to be replicated in larger sets of tasks at varying levels of complexity. Some promising candidates include number puzzles (24, four fours, or Sudoku), Cryptoarithmic problems, and synthetic number problems. Alternatively, problems may be chosen based on their computational complexity (e.g., NP-Hard class of problems). Moreover, we examined only a few selected LLMs among the vast array available today. Even across variants of the same models, the agents can differ drastically in abilities and performance. Therefore, future studies need to examine a larger, diverse pool of models. Finally, as complex problems allow numerous solving strategies, the paths from input information to output performance need to be traced for causal attribution. In this pursuit, XAI methods such as MI reflect the cognitive models used to explain human behavior as promising ways to explain AI reasoning from its elementary processes. Our study highlights several common mistakes, such as producing non-24 outcomes, breaking 24 game rules (e.g., reusing numbers or omitting some), and the breakdowns of reasoning chains, that MI could unpack by identifying the building blocks (e.g., features, circuits, and motifs) involved in reasoning and examining how these internal representations drift during extended reasoning.

7 Conclusions

Our examination of the mathematical reasoning abilities of LLM-based agents revealed clear strengths in deterministic operations such as basic arithmetic, advanced calculations, and prime number checks. However, performance dropped sharply on tasks requiring trial-and-error reasoning, such as the 24 game. These outcomes underscore a significant gap between human and LLM performance in managing uncertainty and solving complex problems. More generally, our study highlights the merits of conducting targeted experiments to systematically evaluate and explain LLM abilities in navigating complex spaces. Further explorations of this nature may pave the way for improved AI safety by revealing ways to improve large LLM reasoning, as well as by drawing clearer distinctions between AI and human reasoning in complex real-world problems.